# Immersion

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## 1. Immersion

Immersion is a transfer (shift) of perception from physical to virtual world.

Immersion = User + Device + Attractive Scenario + Digital 3D Creation

Each component from this equation plays a slightly different role:

- **The user** → He brings attention, perception, prior knowledge, and willingness to engage;
- **The device** → It enables sensory input/output (VR headset, AR glasses, haptic devices, etc.);
- **The attractive scenario (Narrative/Context)** → It provides meaning and motivation to stay engaged;
- **The digital 3D creation (Environment/Assets)** → It provides visual, auditory, and interactive world.

The ***immersion state*** is not the sum of these factors, but the **synergy** among them:

➔ If the user is **distracted**, immersion breaks, even with high-quality tech.
➔ If the device is **poor**, presence is reduced, even if the scenario is rich.
➔ If the scenario **lacks engagement**, the user may remain detached.
➔ If the digital 3D world isn't **coherent** or **realistic**, the illusion collapses.

**Immersion** is the psychological state that emerges when a motivated user interacts with a digital environment through appropriate devices, within a coherent and engaging scenario.

**Immersion** can be understood as the transfer of perception from the physical world to the virtual world. It is implemented through two complementary sides:

1. **Mental and emotional engagement:** Which reinforces the user's psychological adherence to the experience. **Mental engagement** is built by cognitive challenge and meaningful tasks, while emotional engagement is built by narrative, aesthetics, and affective triggers. Together, they make the VR world feel compelling and "real" to the mind.
2. **Sensory stimulation:** It is achieved through high-quality hardware and precise 3D modeling that create a realistic perceptual environment. Sensory stimulation is obtained by delivering coherent, high-quality, and synchronized multisensory inputs. This tricks the brain into perceiving the virtual as real, complementing the mental and emotional engagement.

The combination of these two sides leads to a state of total immersion.

## 2. Taxonomy of VR Interfaces

### 1 Large Consumer Interfaces (mainstream, widely adopted)

◇ **Head-Mounted Displays (HMDs):** Wearable display devices that cover the eyes, presenting stereoscopic 3D images and often integrating head-tracking sensors. Its functioning principle is as follows:

➔ Two small screens (or one split screen) project slightly different images to each eye → stereoscopy.
➔ Gyroscopes, accelerometers, and external cameras track head orientation and movement.
➔ The displayed image updates in real time according to the user's gaze direction, producing a sense of presence.

**Examples**: Meta Quest, HTC Vive, PSVR, Apple Vision Pro.

Available on zenodo – DOI : 10.5281/zenodo.17251375

◇ VR Boxes / Mobile VR Shells: Low-cost headsets (like Google Cardboard) that use a smartphone as the display and processor. Its functioning principle is as follows:

- ➔ A smartphone screen displays split stereoscopic images.
- ➔ Lenses inside the box focus the image for each eye.
- ➔ Basic head tracking through the phone's gyroscope and accelerometer.

**Examples:** Google Cardboard, Samsung Gear VR.

**2 Niche Interfaces (specialized, high-cost, or limited adoption)**

◇ Data Gloves (Haptic or Tracking Gloves): Wearable gloves equipped with sensors to capture hand/finger movement and sometimes provide haptic or force feedback. Its functioning principle is as follows:

- ➔ **Tracking**: flex sensors, IMUs, or magnetic sensors measure finger bending and hand orientation.
- ➔ **Feedback**: vibrotactile motors or pneumatic/force systems provide sensations of touch or resistance.
- ➔ Input is translated into gestures or manipulation of virtual objects.

**Examples:** Manus VR, HaptX, SenseGlove.

◇ Immersive Desktops (Fish Tank VR, 3D monitors): Desktop-based stereoscopic displays that give a 3D impression, sometimes combined with motion tracking. Its functioning principle is as follows:

- ➔ A stereoscopic monitor (glasses-based or glasses-free) presents two slightly offset images.
- ➔ Head/eye tracking adjusts perspective dynamically.
- ➔ The user perceives a "window" into the virtual world.

**Examples:** zSpace, early 3D CAD systems.

◇ CAVEs (Cave Automatic Virtual Environments): Room-sized systems with projection screens on multiple walls, floor, and ceiling. Users wear stereoscopic glasses. Its functioning principle is as follows:

- ➔ Projectors display synchronized stereoscopic images on walls/floor.
- ➔ Tracking systems adjust images based on user position.
- ➔ Multiple users can share the same immersive space, though usually one "drives" the perspective.

**Examples:** CAVE2 (EVL, University of Illinois), large industrial visualization rooms.

**3 Auxiliary Interfaces (immersion enhancers, not standalone)**

◇ Surround and 3D Sound Systems: Audio systems that reproduce spatialized sound, giving the impression that sounds originate from specific directions in 3D space. Its functioning principle is as follows:

- ➔ **Binaural render**ing simulates how sound waves reach each ear with slight timing/volume differences.
- ➔ **Ambisonics or surround** systems reproduce sounds in 360° around the listener.
- ➔ Coupled with head tracking, sounds shift dynamically as the user moves.

◇ Sensor Systems (Motion Detection and Tracking): Devices that capture the position, orientation, and movement of the user's body, head, hands, or eyes. Its functioning principle is as follows:

- ➔ **Optical tracking**: cameras (e.g., inside-out tracking in HMDs, or external sensors like SteamVR base stations) follow markers or natural features.
- ➔ **Inertial tracking**: accelerometers and gyroscopes detect movement and orientation.
- ➔ **Hybrid systems**: combine optical + inertial for accuracy.
- ➔ Data is fed into the VR system to update the user's perspective and interactions in real time.

**Examples:** Oculus Insight (inside-out), SteamVR Lighthouse, eye-tracking modules.

◇ Haptic Accessories (optional but increasingly popular): Wearable or environmental devices that provide touch, vibration, or force feedback to simulate physical sensations. Its functioning principle is as follows:

- ➔ Vibrational motors, pneumatic bladders, or force-feedback actuators generate physical sensations.
- ➔ Devices can simulate textures, impacts, resistance, or even full-body effects.

**Examples:** bHaptics vest, Teslasuit, omnidirectional treadmills (Virtuix Omni).

## 3. Behavior and meaning in VR

In virtual reality, interaction relies on a **synchronized bidirectional communication** between the user and the virtual environment. This interaction can be described through four complementary concepts:

### 3.1. User-to-System Flow (Input)

- ➔ The user generates **behavioral inp**ut through actions, gestures, or decisions.
- ➔ These inputs are captured by devices (e.g., sensors, controllers, gloves) and translated into machine-readable commands.

### 3.2. System-to-User Flow (Output)

- ➔ The virtual environment delivers **sensory output** such as visual, auditory, and haptic data.
- ➔ These outputs inform, guide, and influence the user's perception and subsequent actions.

### 3.3. Interpretation (Coding)

- ➔ The **user interprets** the system's outputs and adjusts behavior accordingly.
- ➔ The **system interprets** the user's inputs and updates the environment dynamically.
- ➔ Meaning emerges through this mutual interpretation.

### 3.4. Synchronized Communication

- ➔ The exchange is not merely two-way but **time-sensitive and coordinated**:
  - ✓ Inputs and outputs must be processed and responded to in real time.
  - ✓ Any delay or desynchronization (e.g., latency, lag) disrupts immersion and presence.
- ➔ Synchronization ensures that user actions and system feedback form a coherent loop.

### 3.5. Communication Scenario (Organizing Framework)

- ➔ This bidirectional synchronized communication is structured by a **communication scenario**:
  - ✓ A framework of **rules of interaction** (how actions are recognized and responded to).
  - ✓ A **semantic layer** (how signals, data, and feedback acquire meaning).
  - ✓ A set of **goals or narratives** (what the user seeks to achieve within the environment).
- ➔ The scenario guarantees that both behavior and feedback contribute to an intelligible, meaningful, and immersive experience.

## 4. Living in VR

Transitioning between the real and virtual worlds involves cognitive adaptation processes, both upon entering a VR environment and upon returning to reality, which are essential to understanding immersion and presence:

### Entering a VR world (adaptation)

- ➢ When users first enter a VR environment, they go through a **short cognitive and perceptual adaptation period** (seconds to minutes).
- ➢ During this phase, they learn the **rules of interaction and communication**, understand the **scenario's logic**, and align their **sensorimotor expectations** with the virtual environment.

### Living in the VR world (immersion & presence)

- ➢ Once these rules are understood, the user experiences **immersion** (the system's technical capacity to isolate the senses) and, more importantly, **presence** (the psychological feeling of "being there").
- ➢ At this point, cognition is effectively **oriented toward the VR world**, as attention, perception, and decision-making processes adapt to the virtual context.

### Leaving the VR world (readaptation)

- ➢ On exiting VR, the user often needs a **readaptation period** to recalibrate perception and cognition to the real environment.
- ➢ This is related to **aftereffects** such as disorientation, perceptual lag, or reduced motor accuracy, which are well documented in VR studies.

The need for cognitive adaptation when entering and leaving VR reflects the depth of immersion and presence. If a user feels no difference between both transitions, it may indicate either a shallow level of immersion, or conversely, an extremely seamless VR experience.